

AI4Trade Global Competition

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Flow of the presentation



Brief on Challenge – as I understood it



Data preparation



Feature engineering



Modelling methodology

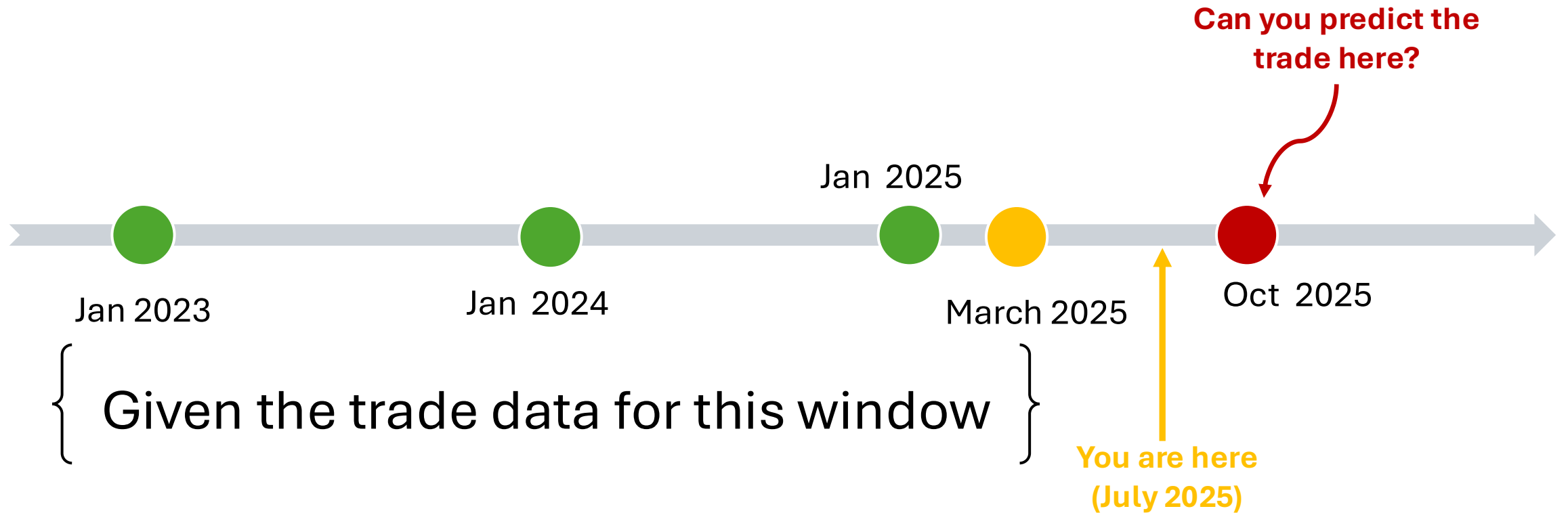


Ensemble strategy



Conclusion

Challenge statement



Challenge Definition & Evaluation

- **What is evaluated?**

Accuracy of **October 2025** bilateral trade forecasts

At **HS-4** product level, for **exports and imports**, for the **USA and China**

- **How is performance measured?**

sMAPE (symmetric mean absolute percentage error)

Evaluated **against official trade data when released**

- **Minimum acceptance criterion**

Forecasts must **outperform a naïve baseline**

(simply carrying forward the latest available month)

The gap plugging



Downloaded monthly data from www.trademap.org for selected partners

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Data exploration

- Is the dataset zero dense/sparse?
- What's the export and import pattern?
- Are there seasonal effects?
- Is there something missing in the fields?
- What happens when we aggregate at national level?
- Does it pass the smell test visually when we plot monthly data?

Aggregated data at national level

Dataset	Number of rows	Number of columns	Unique HS4 lines	Unique HS6 lines	Unique partner countries
CHN 2023	4.4 million	7	1220	5391	249
CHN 2024	4.4 million	7	1219	5378	248
USA 2023	4.2 million	7	1230	5583	229
USA 2024	4.2 million	7	1229	5583	229

Final seven columns: **[origin, destination, hs6, hs4, trade_flow, month, value]**

Decision: The modelling and prediction will be at hs6 level, aggregated to hs4 during submission. No filtering out any hs6/hs4 to make the model generic and transferable to policy applications.

Do we need so many partners and all HS4 for this competition?

Picking partners

- **Four datasets** for CHN/USA and trade_flow pairs.
- Top 30 partners for trade flows were retained - > **led to significant row reduction, zooming into the problem space**
- Based on the clean up and selection, **around 9.17 million rows** of total data was retained as shown in the table below.
- **The data covered the period from Jan 2023 to July 2025 for USA and August 2025 for CHN.**

origin	trade_flow	row_count	unique_hs6_count	unique_hs4_count	Partners
CHN	Export	3059124	5283	1217	30
CHN	Import	1261279	5118	1199	29
USA	Export	2605360	5559	1228	30
USA	Import	2247491	5534	1228	30

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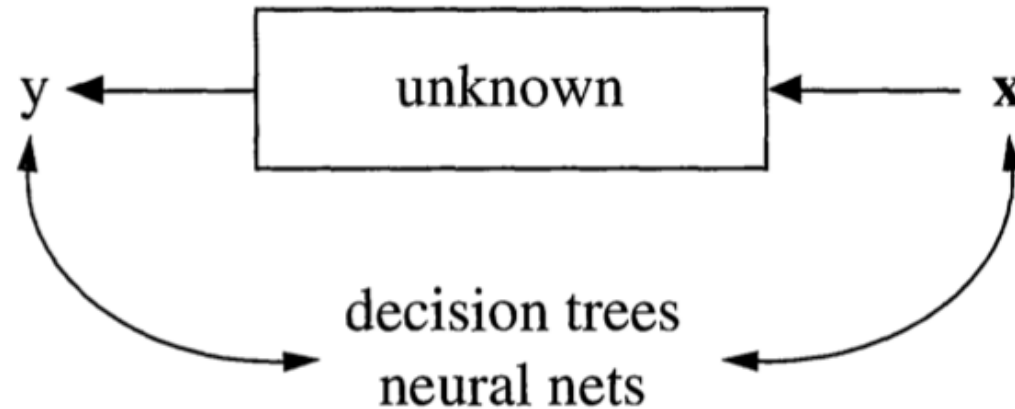
Feature Engineering

- What could be the simplest possible approach in terms of modelling the features?
- What's the least number of features that I can add, while not leaving something important?
- What other things matter for prediction?
 - GDP growth
 - Exchange rate
 - Tariff and non-tariff changes
 - Shipping cost and supply chain dynamics
 - Geopolitics
 - FTAs
 - Unknown unknowns

Leo Breiman's second culture way

The Algorithmic Modeling Culture

The analysis in this culture considers the inside of the box complex and unknown. Their approach is to find a function $f(\mathbf{x})$ —an algorithm that operates on $\mathbf{x}$ to predict the responses $\mathbf{y}$. Their black box looks like this:



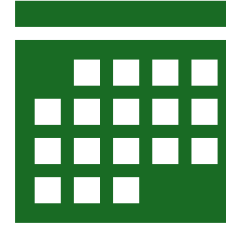
From: Statistical modelling: The two cultures by Leo Breiman

Feature design principles



Principles

Economically Interpretable
Lag safe by construction
Robust to sparsity and zeros
Segment and horizon aware



Feature Buckets

Temporal / Calendar
Lagged Levels & Memory
Trend & Volatility
Cross-Flow Coupling
Macro Signals
Activity & Zero Dynamics

Cross-flow signals

- **Trade flows are interdependent**
Imports and exports of the same product often move together through supply-chain linkages.
- **Cross-flow features capture this linkage**
Export models observe lagged import behavior (and vice versa) for the same product group.
- **Adds information beyond own-history lags**
Improves predictions when trade shifts are driven by sourcing, processing, or re-export dynamics.

Macro Signals: Global Supply Chains & Business Cycles

Global Supply Chain Pressure Index (GSCPI)

Captures worldwide logistics stress — shipping delays, freight costs, and bottlenecks. Data available from NY Fed.

Composite Leading Indicator (CLI)

Signals turning points in global economic activity ahead of observed trade flows. Data available from St. Louis Fed.

Used as lagged, smoothed signals

Both indices are lag-shifted and averaged before use to ensure strict temporal validity.

Sl no.	Feature	Family (Bucket)	Computation (summary)	Why it helps
1	roll_std6	Trend & Volatility	$\text{std}(y[t-6..t-1])$	Uncertainty
2	pctchg1	Trend & Volatility	$(y(t-1)-y(t-2))/y(t-2)$	MoM momentum
3	pctchg3	Trend & Volatility	$(y(t-1)-y(t-4))/y(t-4)$	3-month momentum
4	ma3	Trend & Volatility	$\text{mean}(y[t-3..t-1])$	Noise reduction
5	ma6	Trend & Volatility	$\text{mean}(y[t-6..t-1])$	Stability
6	ma12	Trend & Volatility	$\text{mean}(y[t-12..t-1])$	Baseline
7	yoy_delta	Trend & Volatility	$y(t-1) - y(t-13)$	YOY momentum
8	weight_sqrt_ma12	Training Stabilization	$\sqrt{(\text{ma12} + 1)}$	Stabilize fitting
9	month_num	Temporal & Calendar	Month integer 1–12	Seasonality
10	quarter	Temporal & Calendar	Quarter 1–4	Quarterly cycles
11	month_id	Temporal & Calendar	Monotone month index	Trend / drift
12	month_sin	Temporal & Calendar	$\sin(2\pi \cdot \text{month_num}/12)$	Cyclic pattern
13	month_cos	Temporal & Calendar	$\cos(2\pi \cdot \text{month_num}/12)$	Cyclic pattern
14	gscpi_ma2_shift	Macro Drift	MA2 of GSCPI, shifted +1 (CHN) / +2 (USA)	Logistics pressure
15	oecd_cli_ma2_shift	Macro Drift	MA2 of CLI, shifted +1 (CHN) / +2 (USA)	Turning points
16	origin_total_exports_ma3	Macro Drift	MA3 of exports (origin)	Aggregate momentum
17	origin_total_imports_ma3	Macro Drift	MA3 of imports (origin)	Intermediate-goods demand
18	lag1	Lagged Levels & Memory	$y(t-1)$	Contract inertia
19	lag2	Lagged Levels & Memory	$y(t-2)$	Near-term memory (CHN h=2)
20	lag3	Lagged Levels & Memory	$y(t-3)$	Near-term memory (USA h=3)
21	lag6	Lagged Levels & Memory	$y(t-6)$	Mid-term memory
22	lag10 / lag9	Lagged Levels & Memory	$y(t-10)$ CHN / $y(t-9)$ USA	Horizon-tuned memory
23	lag12	Lagged Levels & Memory	$y(t-12)$	YOY seasonality
24	cross_flow_lag1	Cross-Flow Coupling	Opposite flow t–1 at HS-4	Complement / substitute
25	cross_flow_ma3	Cross-Flow Coupling	MA3 of opposite flow	Smoother co-movement
26	cross_flow_lag13	Cross-Flow Coupling	Opposite flow t–13	YOY coupling
27	hs_section_prefix	Context / Structural	First 2–4 digits of HS	Taxonomy signal
28	partner_tier_rank	Context / Structural	Rank by trailing 12-m flow	Market depth
29	zero_share_flag	Context / Structural	Near-zero share recently	Suppress spurious fits
30	was_trade_lag1	Activity & Zero Dynamics	$1\{y(t-1) > 0\}$	Extensive margin
31	consec_zero_run	Activity & Zero Dynamics	Run length of zeros up to t–1	Entry / exit dynamics

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Framing the Modelling Problem

Modelled as a *direct multi-step forecasting* problem, not a one-step regression task

Design principles

- Target is $y_{\text{target}} = y(t + h)$ within each series (origin, destination, hs6, trade_flow).
- CHN Export/Import ($h=2$), USA Export/Import ($h=3$)
- China: available data up to Aug 2025 → **test context @ Aug 2025** → predict Oct 2025.
- USA: available data up to Jul 2025 → **test context @ Jul 2025** → predict Oct 2025.
- Train rows = months t where $y(t+h)$ exists in data (non-null).
- **No lookahead**: all features were built using data $\leq t-1$
- **No pruning of HS6 series.**

Model Search Space & Empirical Filtering

Tree-based models (retained)

- XGBoost – Tweedie
- XGBoost – Log1p
- LightGBM – Log-RMSE

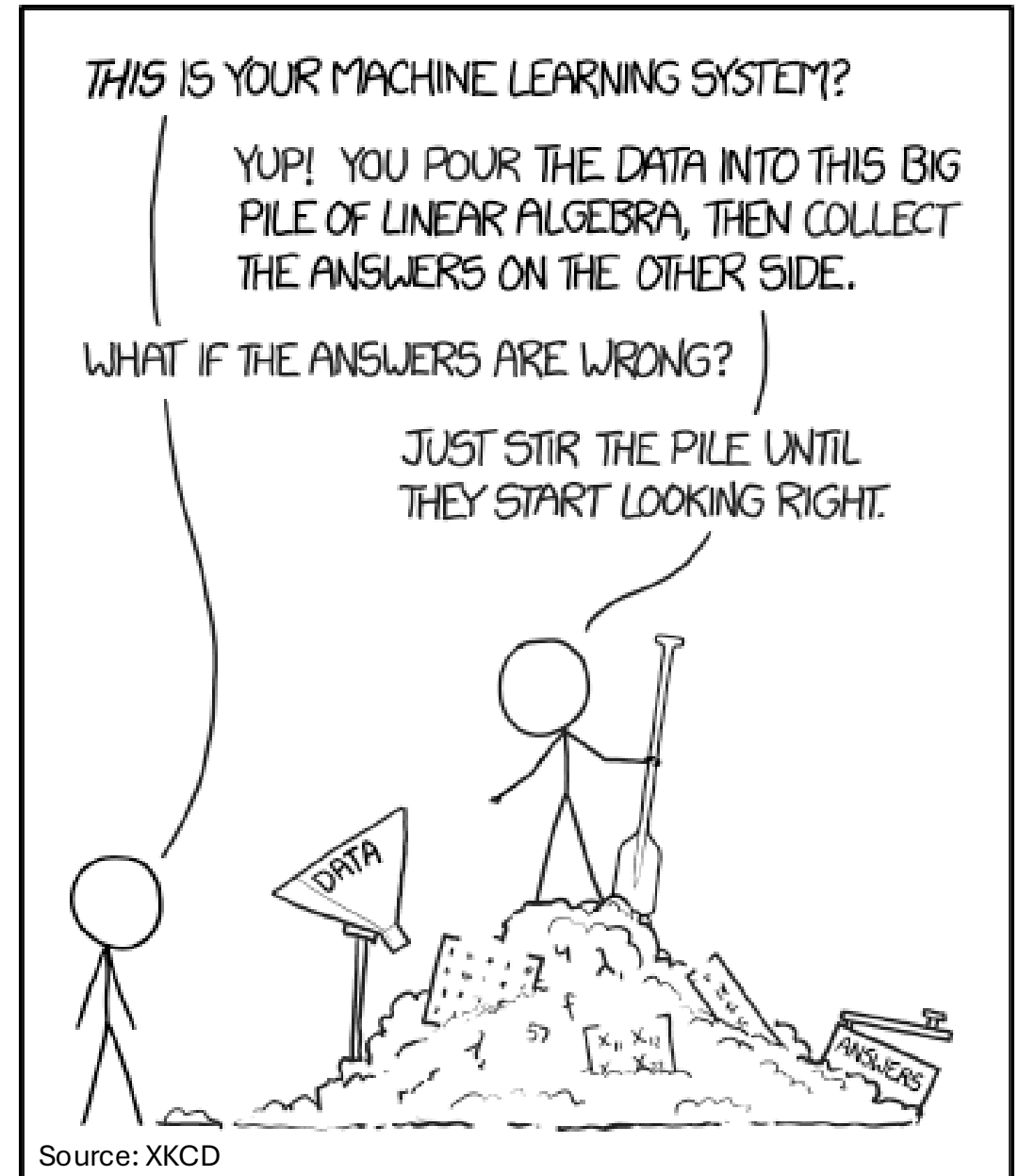
Neural / global models (tested, discarded)

- N-HiTS
- N-BEATSx / TFT

Classical baselines (benchmarks, not used)

- Seasonal Naïve
- STL-ETS / SARIMAX-lite

Only models that consistently beat the naïve baseline under sMAPE at HS-4 aggregation were retained.



Final list

Model	What it optimizes	Importance
XGBoost – Tweedie	Zero-inflated, heavy-tailed loss	Handles no-trade months and extreme spikes naturally
XGBoost – Log1p	Squared error on log scale	Stabilizes learning on mid-range, regular trade flows
LightGBM – Log-RMSE	Simple, low-variance learner	Adds robustness and prevents over-reaction to noise

These models are usual competition winners

Segmented Modelling approach

Trade dynamics differ by country and flow

Export and import behavior varies structurally between the USA and China. Even some partners change.

Four distinct modeling segments

- China – Exports
- China – Imports
- USA – Exports
- USA – Imports

Segment-specific learning improves accuracy

Allows models to adapt to different seasonality, volatility, and partner structures.

Each segment is modeled independently and later combined into a unified forecast.

Segment-Specific Forecast Horizons and Cut-Offs

	FLOW	HORIZON	LAST_CONTEXT	MAX_TRAIN	PREDICT
CHN	Export	2	2025-08-01	2025-06-01	2025-10-01
CHN	Import	2	2025-08-01	2025-06-01	2025-10-01
USA	Export	3	2025-07-01	2025-04-01	2025-10-01
USA	Import	3	2025-07-01	2025-04-01	2025-10-01

Cross Validation Strategy (Time-Aware & Horizon-Matched)

Rolling, time-ordered folds

Models are always trained on the past and validated on the future — no random splits.

Horizon-matched validation

Validation targets match the final task
($h = 2$ for China, $h = 3$ for USA).

Later folds weighted more heavily

Performance closer to October 2025 matters more for model selection and ensembling.

Validation is done on HS-4 aggregates using sMAPE, compared against a naïve carry-forward baseline.

Cross Validation splits - China

Fold	train_end	gap_start	val_target	weight
C1	2024-05-01	2024-06-01	2024-08-01	1.00
C2	2024-06-01	2024-07-01	2024-09-01	1.00
C3	2024-07-01	2024-08-01	2024-10-01	1.25
C4	2024-09-01	2024-10-01	2024-12-01	1.00
C5	2025-03-01	2025-04-01	2025-06-01	1.50
C6	2025-04-01	2025-05-01	2025-07-01	1.50

CV scores for one of the models – LGBM

Fold	train_end	val_target	wt.	hs4_smape	weighted
C1	01/05/24	01/08/24	1	0.471	0.471
C2	01/06/24	01/09/24	1	0.461	0.461
C3	01/07/24	01/10/24	1.25	0.480	0.600
C4	01/09/24	01/12/24	1	0.476	0.476
C5	01/03/25	01/06/25	1.5	0.424	0.636
C6	01/04/25	01/07/25	1.5	0.415	0.622

Weighted mean HS-4 sMAPE across folds: 0.451

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Ensemble & Final Forecast Construction

- **Model performance evaluated via time-based CV**
Each of the 12 models is scored using horizon-matched, rolling validation folds.
- **Weights derived from validation accuracy**
Models that perform better under sMAPE receive higher weight.
- **Weighted ensemble for robustness**
Final forecasts are a weighted combination of complementary models, reducing reliance on any single learner.

Blending is done within each segment and then aggregated for final submission.

Why a Weighted Ensemble?

- **No single model is reliable everywhere**
Different models succeed under different trade regimes and shock conditions.
- **Blending spreads model risk**
Reliance on one model risks overfitting to specific patterns or periods.
- **Errors partially cancel out**
Model-specific biases differ, so combining them reduces overall error.
- **Weights reflect empirical reliability**
Cross-validation assigns more influence to models that generalize better.
- **The ensemble trades peak performance for stability and generalization.**

How is the ensemble built?

Step 1: Segment-level evaluation

- For each segment (CHN/USA × Export/Import), generate **OOF predictions**
- Compute **fold-weighted HS-4 sMAPE** per model
- Later folds receive higher weight

Step 2: Weight calculation

- Base weights = **inverse sMAPE**
- Regularized with prior
(0.45 Tweedie / 0.25 Log1p / 0.30 LGBM)
- Final weight = **0.70 × data-driven + 0.30 × prior**
- **Recency tilt**: +10% if model wins latest folds
- **Stability**: weights clipped to **[0.15, 0.65]**, normalized

Step 3: Forecast construction

- Segment-level weighted average of model forecasts
- Segments combined only after aggregation

Final Forecast = $\sum (\text{model weight} \times \text{model prediction})$

Validation (2024H2–2025H1) showed ensemble HS4 sMAPE≈0.77 vs naive≈0.94 at the time of submission

Model Performance and Ensemble Weights(Validation-Based)

Segment	Model	Weighted CV sMAPE	Final Ensemble Weight
CHN – Export	XGB-Tweedie	0.459	0.398
	LGBM-LogRMSE	0.478	0.309
	XGB-Log1p	0.481	0.293
CHN – Import	LGBM-LogRMSE	0.625	0.349
	XGB-Log1p	0.632	0.300
	XGB-Tweedie	0.652	0.351
USA – Export	XGB-Tweedie	0.933	0.394
	LGBM-LogRMSE	1.066	0.317
	XGB-Log1p	1.134	0.288
USA – Import	XGB-Tweedie	0.871	0.389
	LGBM-LogRMSE	0.953	0.322
	XGB-Log1p	1.028	0.290

Post processing and submission

Post-processing

- Forecasts generated at **HS-6** level
- Aggregated to **HS-4** for evaluation & submission
- Values rounded to **USD**

Submission rules applied

- Retained **top 20 partner countries** per (*origin, trade_flow*)
- Enforced **≥ 200 HS-4 lines** per partner
- One row per (*Country1, Country2, HS-4, Flow*) with values in *USD*
- Submission file generated

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Key takeaway & Code repository



By constraining the problem with time, structure, and evidence, the model is forced to generalize.



Code accessible at this github link/QR:

<https://tinyurl.com/ai4trade>



Thank you
